

BYRON BAY, NU, Canada

WMO: 719290

Lat: 68.750N

Lon: 109.070W

Elev: 302

StdP: 14.54

Time Zone: -7.00 (NAM)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.7	-40.3	-39.5	0.6	-32.8	-38.3	0.6	-32.4	N/A	N/A	N/A	N/A	N/A	N/A	1.113

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	63.0	54.6	59.8	52.4	56.7	50.9	55.6	61.7	53.2	58.6	51.1	56.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.1	58.5	59.6	49.7	53.4	56.5	47.5	49.1	54.3	23.7	62.1	22.2	58.8	21.0	56.2	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-45.8	69.7	2.3	3.7	-47.5	72.3	-48.8	74.5	-50.1	76.5	-51.8	79.2
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-26.2	-24.8	-18.3	-4.7	16.0	38.0	49.1	43.9	32.1	13.3	-10.9	-19.1
	DBStd	28.61	9.74	11.16	9.95	10.85	10.33	8.60	5.88	6.45	6.73	10.79	11.52	10.10
	HDD50	15594	2362	2095	2117	1642	1054	372	91	215	537	1138	1828	2143
	HDD65	20969	2827	2515	2582	2092	1519	810	493	655	987	1603	2278	2608
	CDD50	100	0	0	0	0	0	12	64	24	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	4.7	0.1	0.1	0.1	0.2	0.2	0.4	1.0	1.0	0.7	0.6	0.2	0.1
	PrecMax	8.0	1.3	0.7	0.8	0.6	0.9	3.0	2.8	2.4	3.1	1.1	0.4	1.1
	PrecMin	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.0
	PrecStd	1.5	0.2	0.2	0.2	0.2	0.2	0.6	0.7	0.6	0.6	0.3	0.1	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	11.9	12.9	25.9	37.8	63.7	70.2	65.6	49.5	37.8	23.3	15.5
		MCWB	9.8	11.4	12.0	24.8	36.2	52.5	59.5	56.5	46.0	37.3	22.8	15.1
	2%	DB	2.1	1.6	6.5	20.5	34.4	58.6	64.5	61.1	46.0	33.5	18.0	7.5
		MCWB	1.4	0.8	5.6	19.7	33.6	50.7	55.1	54.6	44.2	32.9	17.4	6.9
	5%	DB	-5.0	-3.1	1.5	16.3	32.3	54.4	61.4	57.2	43.5	30.8	12.4	3.0
		MCWB	-5.6	-3.8	0.9	15.6	31.8	48.8	53.4	52.4	41.9	30.3	11.6	2.5
	10%	DB	-12.4	-9.2	-3.5	11.3	30.3	50.4	58.7	53.6	40.7	28.0	6.9	-3.4
		MCWB	-12.8	-9.5	-4.0	10.6	29.9	45.8	51.7	49.1	39.4	27.4	6.3	-4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	12.0	12.5	25.0	36.8	54.0	60.6	58.0	46.7	37.5	23.0	15.2
		MCDB	11.3	12.5	13.3	25.7	37.8	60.3	68.4	64.3	48.1	38.0	23.4	15.9
	2%	WB	3.2	5.8	6.4	19.7	33.6	51.2	56.6	55.1	44.1	32.8	17.3	7.0
		MCDB	3.7	6.4	7.0	20.4	34.2	57.5	63.0	60.2	45.7	33.2	17.9	7.5
	5%	WB	-1.6	-1.2	1.6	15.5	32.0	48.8	54.1	52.7	42.0	30.4	11.7	2.8
		MCDB	-0.9	-0.6	2.3	16.3	32.4	54.7	60.2	57.0	43.4	30.7	12.4	3.4
	10%	WB	-9.1	-5.4	-3.1	10.7	29.9	45.9	52.1	49.6	39.5	27.5	6.3	-3.3
		MCDB	-8.6	-4.7	-2.6	11.4	30.3	50.2	57.6	53.0	40.6	28.0	6.9	-2.9
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	10.0	11.3	9.3	8.4	10.5	8.6	6.1	8.7	8.8	9.0
		MCDBR	17.1	14.7	14.0	13.2	8.1	13.6	14.8	13.5	8.1	7.8	10.2	12.9
		MCWBR	16.9	13.9	13.8	13.0	8.0	9.4	8.8	8.7	7.0	7.9	10.2	12.8
	5% WB	MCDBR	17.0	14.5	14.2	13.1	8.2	13.5	14.3	13.2	8.0	7.7	10.4	13.0
		MCWBR	17.0	14.3	14.0	12.9	8.1	9.7	9.8	9.3	7.4	7.9	10.3	12.9
Clear-Sky Solar Irradiance	taub	0.146	0.206	0.242	0.289	0.318	0.343	0.356	0.335	0.292	0.230	0.158	N/A	
	taud	1.901	2.073	2.227	2.142	2.144	2.312	2.386	2.475	2.505	2.193	1.901	N/A	
	Ebn at Noon	33	189	256	270	274	270	262	256	240	174	21	0	
	Edh at Noon	2	14	23	34	38	34	31	25	18	12	2	0	
All-Sky Solar Radiation	RadAvg	7	142	661	1362	2010	1997	1714	1109	530	169	21	0	
	RadStd	1	16	52	95	157	107	118	98	60	26	2	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page